CHAPTER 1 FUNCTION AND LIMIT

Function: Function is a rule that assigns each value of set to a unique value of another set, while a relation is a map between two non-empty sets which may or may not be a function

A relation can be checked either it's a function or not graphically

Vertical Line Test: Draw the graph of a given relation and then draw vertical lines if each vertical line intersects the graph of relation at only one point then this relation will be a function otherwise not a function

Horizontal Line Test: Horizontal lines are drawn to check whether a function is onto or not in similar sense of vertical line test

Even Function: A function $f(x)$ is said to be an even function if $f(-x) = f(x)$, note that graph of even function is symmetric about y-axis.

Odd Function: A function $f(x)$ is said to be an even function if $f(-x) = -f(x)$, note that graph of odd function is not symmetric about y-axis.

Laws of Limit	
$\lim\limits_{x\to a}\{f(x) \pm g(x)\} = \lim\limits_{x\to a} f(x) \pm \lim\limits_{x\to a} g(x)$	$\lim\limits_{x\to a}\{f(x) \cdot g(x)\} = \lim\limits_{x\to a} f(x) \cdot \lim\limits_{x\to a} g(x)$
$\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim\limits_{x\to a} f(x)}{\lim\limits_{x\to a} g(x)}$	$\lim_{x\to a}\{f(x)\}^n = \{\lim_{x\to a} f(x)\}^n$ where n is constant
$\lim\limits_{x\to a} Af(x) = A \lim\limits_{x\to a} f(x)$	$\lim\limits_{x\to a}\{f(x) + A\} = \lim\limits_{x\to a} f(x) + A$

Formulas of Limit		
$\lim\limits_{x\to 0} \dfrac{\sin\theta}{\theta} = 1$	$\lim\limits_{n\to\infty} \left(1 + \dfrac{1}{n}\right)^n = e$	$\lim\limits_{x\to a} \dfrac{x^n - a^n}{x - a} = na^{n-1}$
$\lim\limits_{x\to 0} \dfrac{\tan\theta}{\theta} = 1$	$\lim\limits_{x\to 0}(1 + x)^{\frac{1}{x}} = e$	$\lim\limits_{x\to 0} \dfrac{a^x - 1}{x} = lna$

Formulas of Hyperbolic functions		
$\sinh\theta = \dfrac{e^\theta - e^{-\theta}}{2}$	$\cosh\theta = \dfrac{e^\theta + e^{-\theta}}{2}$	$\tanh\theta = \dfrac{e^\theta - e^{-\theta}}{e^\theta + e^{-\theta}}$
$\text{cosech}\,\theta = \dfrac{2}{e^\theta - e^{-\theta}}$	$\text{sech}\,\theta = \dfrac{2}{e^\theta + e^{-\theta}}$	$\coth\theta = \dfrac{e^\theta + e^{-\theta}}{e^\theta - e^{-\theta}}$

Existence of Limit: Limit of a function $f(x)$ is said to exist at $x = a$, if

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$$

$Lf(x) = \lim\limits_{x\to a^-} f(x)$ is known as left limit and $Rf(x) = \lim_{x\to a^+} f(x)$ is known as right limit

Continues Function: A function$f(x)$ is said to be continues function at $x = a$, if

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$$

OR

$$\lim_{x\to a} f(x) = f(a)$$

CHAPTER 2 DIFFERENTIATION

Derivative is the rate of change of one variable w.r.t other variable at a point is called derivative Mathematically,

$$f'(x)=\lim_{\delta x\to 0}\frac{f(x+\delta x)-f(x)}{\delta x} \qquad OR \qquad \lim_{\delta x\to 0}\frac{\delta y}{\delta x}=\frac{dy}{dx}$$

Example:

- Derivative of displacement w.r.t time gives velocity
- Derivative of velocity w.r.t time gives acceleration
- Derivative of momentum w.r.t time gives force
- Derivative of angular momentum w.r.t time gives torque

DERIVATIVE OF TRIGONOMETRIC FUNCTIONS		
$\frac{d}{dx}\sin x=\cos x$	$\frac{d}{dx}\cos x=-\sin x$	$\frac{d}{dx}\tan x=sec^2 x$
$\frac{d}{dx}\cot x=-cosec^2 x$	$\frac{d}{dx}\sec x=\sec x\tan x$	$\frac{d}{dx}\mathrm{cosec}\, x=-\mathrm{cosec}\, x\cot x$
DERIVATIVE OF INVERSE TRIGONOMETRIC FUNCTIONS		
$\frac{d}{dx}\sin^{-1}x=\frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}\cos^{-1}x=-\frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}\tan^{-1}x=\frac{1}{1+x^2}$
$\frac{d}{dx}\sec^{-1}x=\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$	$\frac{d}{dx}\mathrm{cosec}^{-1}x=-\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$	$\frac{d}{dx}\cot^{-1}x=-\frac{1}{1+x^2}$
DERIVATIVE OF HYPERBOLIC FUNCTIONS		
$\frac{d}{dx}\sinh x=\cosh x$	$\frac{d}{dx}\cosh x=\sinh x$	$\frac{d}{dx}\tanh x=\mathrm{sech}^2 x$
$\frac{d}{dx}\mathrm{sech}\, x=-\mathrm{sech}\, x\tanh x$	$\frac{d}{dx}\mathrm{cosech}\, x=-\mathrm{cosech}\, x\coth x$	$\frac{d}{dx}\coth x=-\mathrm{cosech}^2 x$
DERIVATIVE OF INVERSE HYPERBOLIC FUNCTIONS		
$\frac{d}{dx}\sinh^{-1}x=\frac{1}{\sqrt{x^2+1}}$	$\frac{d}{dx}\cosh^{-1}x=\frac{1}{\sqrt{x^2-1}}$	$\frac{d}{dx}\tanh^{-1}x=\frac{1}{1-x^2}$
$\frac{d}{dx}\mathrm{sech}^{-1}x=\frac{-1}{x\sqrt{1-x^2}}$	$\frac{d}{dx}\mathrm{cosech}^{-1}x=\frac{-1}{x\sqrt{x^2+1}}$	$\frac{d}{dx}\coth^{-1}x=-\frac{1}{1-x^2}$
SOME OTHER FORMULAS OF DERIVATIVES		
Power rule	**Product rule**	**Quotient rule**
$\frac{d}{dx}[f(x)]^n=n[f(x)]^n\frac{d}{dx}f(x)$	$\frac{d}{dx}u.v=v.u'+u.v'$	$\frac{d}{dx}\left(\frac{u}{v}\right)=\frac{v.u'-u.v'}{v^2}$
Chain Rule	$\frac{dy}{dx}=\frac{dy}{dt}\cdot\frac{dt}{dx}$	Where x=x(t) & y=y(t)
Exponential function	**Exponential function with base ''e''**	**Logarithmic function**
$\frac{d}{dx}a^{f(x)}=a^{f(x)}\ln\lvert a\rvert.\frac{d}{dx}f(x)$	$\frac{d}{dx}e^{f(x)}=e^{f(x)}.\frac{d}{dx}f(x)$	$\frac{d}{dx}\ln\lvert f(x)\rvert=\frac{1}{f(x)}.\frac{d}{dx}f(x)$

McLaren's series: If function f(x) is such that $f, f', f'', f''',\ldots$ are differentiable at x=0, then	**Taylor's series:** If function f(x) is such that $f, f', f'', f''',\ldots$ are differentiable a point x, then
$f(x)=f(0)+xf'(0)+\frac{x^2}{2!}f''(0)+\frac{x^3}{3!}f'''(0)+\cdots$	$f(x+h)=f(x)+hf'(x)+\frac{h^2}{2!}f''(x)+\frac{h^3}{3!}f'''(x)+\cdots$

FIRST DERIVATIVE TEST FOR INCREASING/DECREASING FUNCTIONS:

i. A continues function f(x) is said to be increasing function on $]a,\ b[$ if $f'(x) > 0, \forall x \in\ ]a,\ b[$

ii. A continues function f(x) is said to be decreasing function on $]a,\ b[$ if $f'(x) < 0, \forall x \in\ ]a,\ b[$

iii. A continues function f(x) is said to be constant function on $]a,\ b[$ if $f'(x) = 0, \forall x \in\ ]a,\ b[$

SECOND DERIVATIVE TEST FOR EXTREME VALUES:

i. A function f(x) is said to possess relative minima at x=a, if $f''(a) > 0$, provided $f'(a)$ =0

ii. A function f(x) is said to possess relative maxima at x=a, if $f''(a) < 0$, provided $f'(a) = 0$

INCREASING FUNCTION:

A function $f(x)$ is said to be increasing on $[a, b]$ if for every pair of points x_1 , $x_2 \in [a, b]$ such

$$x_1 > x_2 => f(x_1) > f(x_2)$$

DECREASING FUNCTION:

A function $f(x)$ is said to be decreasing on $[a, b]$ if for every pair of points x_1 , $x_2 \in [a, b]$ such

$$x_1 > x_2 => f(x_1) < f(x_2)$$

REMARKS:

- A function which is either increasing or decreasing is called monotonic function
- If $y = f(x)$ be any curve then its derivative represents slope of tangent at some point
- A point on a curve where first derivative is zero i.e. $\boldsymbol{f'(x) = 0}$ is called stationary point or critical point
- A point on a curve where second derivative is zero i.e. $\boldsymbol{f''(x) = 0}$ is called point of infection
- First derivative of y w.r.t x is usually denoted by $\boldsymbol{\frac{dy}{dx}, y_1, y'\ or\ Dy}$ where $\boldsymbol{D = \frac{d}{dx}}$
- Second derivative of y w.r.t x is usually denoted by $\boldsymbol{\frac{d^2y}{dx^2}, y_2, y''\ or\ D^2y}$ where $\boldsymbol{D^2 = \frac{d^2}{dx^2}}$ and so on
- An equation which contains derivatives or differentials is called differential equation

CHAPTER 3 INTEGRATION

Integration is the process of finding anti-derivative of a function

$\int \boldsymbol{f(x)dx}$ is called primitive, anti-derivative or integral. $\int$ *is called sign of integral*,

And **dx** is called differential of x.

For functions of one independent variables Integration is of two types

(i). Indefinite integrals, when limit of integration is not defined

(ii). Definite Integrals, when limit of integration is defined

Definite integrals are commonly used for determining area under the curve, while indefinite integrals are used to solve ordinary differential equations.

Rules of Integration

1. **Simplification:** Simplify the given integrant from product, quotient or whole power into sum or difference.
2. **Completing the Derivative:** If simplification is not possible, then try to complete derivative and use the one of the two formulas

$$\int \{\boldsymbol{f(x)}\}^{\boldsymbol{n}} \boldsymbol{f'(x)dx} = \frac{\{\boldsymbol{f(x)}\}^{\boldsymbol{n}}}{\boldsymbol{n+1}} + \boldsymbol{c}, \qquad \boldsymbol{n \neq -1}$$

$$\int \frac{\boldsymbol{f'(x)}}{\boldsymbol{f(x)}} \boldsymbol{dx} \quad = \boldsymbol{ln|f(x)| + c}$$

3. **Integration by Substitution:**

(i).For the form $\int F\big(f(x)\big)f'(x)dx$, put $f(x) = t$ and then proceed.

(ii).For the form $\int F\big(\sqrt{linear}\big)dx$, put $\sqrt{linear} = t$ and then proceed.

(iii).For the form $\int ln\big(f(x)\big)\frac{f'(x)}{f(x)}dx$, put $\ln(f(x)) = t$ and then proceed.

4. **Integration for Existence of Quadratic expressions**

(i).For form $\int \boldsymbol{F(quadratic\ expression\ )x\ dx}$, complete derivative of the quadratic expression and then proceed

(ii).For the form $\int \boldsymbol{Standard\ quadratic\ dx}$, i.e. x is not present in the numerator then use completing square technique the use appropriate formula.

(iii).For the form $\int \boldsymbol{Pure\ quadratic\ dx}$, use substitutions as

(a).For form $\boldsymbol{a^2 - x^2}, put\ \boldsymbol{x = a\sin\theta}$ (b).For the form $\boldsymbol{a^2 + x^2}, put\ \boldsymbol{x = a\tan\theta}$

(c).For the form $\boldsymbol{x^2 - a^2}, put\ \boldsymbol{x = a\sec\theta}$

5. **Integration for Trigonometric Function:**

Trigonometric functions can be integrated by one of the following ways in order

(i).Simplify and use suitable formula from formula chart

(ii). If simplification is not easy then break the expression into two parts as function and its derivative and the

Use (i) of the rule 3.

(iii).If above two rules failed to work then break the trigonometric function into two parts and use by parts integration.

6. **Integration for Algebraic fraction:**

To integrate the integrals of form $\int \frac{P(x)}{Q(x)} dx$, if fraction is improper then use division to make the fraction proper

When the fraction is proper try to factorize denominator if possible and use partial fractions to simplify the integrant otherwise make some different technique.

7. **Integration by Parts:**

$$\int (\mathbf{1^{st}\ function})\,(\mathbf{2^{nd}\ function})\mathbf{dx} = (\mathbf{1^{st} function})\left(\int \mathbf{2^{nd} function}\right) - \int \left(\left(\int \mathbf{2^{nd} function}\right)\left(\frac{\mathbf{d}}{\mathbf{dx}}\mathbf{1^{st}\ function}\right)\right)\mathbf{dx}$$

8. **Definite Integrals:**

If $F(x)$ is primitive or integral of (x) , then

$$\int_a^b \boldsymbol{f(x)dx = F(b) - F(a)}$$

FORMULAS OF INTEGRATION

1.	$\int sinx\ dx = -cosx + c$	2.	$\int cosx\ dx = sinx + c$
3.	$\int tanx\ dx = -ln\lvert cosx\rvert + c$	4.	$\int tanx\ dx = \ln\lvert secx\rvert + c$
5.	$\int cotx\ dx = \ln\lvert sinx\rvert + c$	6.	$\int cotx\ dx = -\ln\lvert cosecx\rvert + c$
7.	$\int secx\ dx = \ln\lvert secx + tanx\rvert + c$	8.	$\int cosecx\ dx = \ln\lvert cosecx - cotx\rvert + c$
9.	$\int secx \tan x\ dx = secx + c$	10.	$\int cosec\ x \cot x\ dx = -cosec\ x\ + c$
11.	$\int sec^2x\ dx = tanx + c$	12.	$\int cosec^2x\ dx = -cotx + c$
13.	$\int 0\ dx = c$	14.	$\int x^n\ dx = \frac{x^{n+1}}{n+1} + c,\quad n \neq -1$
15	$\int \frac{1}{x} dx = \ln\lvert x\rvert + c$	16.	$\int a^x\ dx = \frac{a^x}{\ln\lvert a\rvert} + c$
17.	$\int e^x(f(x) + f'(x))dx = e^x f(x)\ +\ c$	18.	$\int e^{ax}(a\,f(x) + f'(x))dx = e^{ax}f(x)\ +\ c$
19.	$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a}Tan^{-1}\left(\frac{x}{a}\right) + c$	20.	$\int \frac{1}{\sqrt{a^2 - x^2}} dx = sin^{-1}\left(\frac{x}{a}\right) + c$

Definite integrals are usually to determine area under a curve and between x-axis given as follows

$$area = \int_a^b f(x)dx \quad OR \quad area = \int_a^b f(y)dy$$

Indefinite integrals are used usually in solving differential equation for their exact solutions

CHAPTER 4 ANALYTIC GEOMETRY

Distance Formula in Plane

1. Distance between the two points or length of line segment joining the two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in plane is

$$PQ = \sqrt{(x_1 - x_2)^{\,2} + (y_1 - y_2)^{\,2}}$$
$$or$$
$$PQ = \sqrt{(x_2 - x_1)^{\,2} + (y_2 - y_1)^{\,2}}$$

2. Perpendicular distance from the point $P(x_1, y_1)$ to the line $ax + by + c = 0$, is

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Mid-Point Formula

The mid-point of the line segment joining the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in plane is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Ratio Formula

The point which divides the join of $P(x_1, y_1)$ and $Q(x_2, y_2)$ to the ratio $k_1 : k_2$ resp. in plane is

$$\left(\frac{k_1 x_2 + k_2 x_1}{k_1 + k_2}, \frac{k_1 y_2 + k_2 y_1}{k_1 + k_2}\right)$$

Translation of Axes

In rectangular co-ordinate system (x, y) if origin $O(0,0)$ is shifted to $O'(h, k)$ keeping the axes as parallel, then transformation equation for new co-ordinate system $(X,\ Y)$ are given as follows

$$x = X + h, \qquad y = Y + k$$

Rotation of Axes

In rectangular co-ordinate system (x, y) if axes are rotated through an angle $\theta < 90°$, keeping the origin as same, then transformation equations for new co-ordinate system $(X,\ Y)$ are given as follows

$$X = x\ cos\theta + y\ sin\theta, \qquad Y = -x\ sin\theta + y\ cos\theta$$

Note: Inverse transformations can be obtained by interchanging (x, y) by $(X,\ Y)$, $(X,\ Y)$ by (x, y) and θ by $-\theta$

Slope of Line

1. Slope $'m'$ of line joining points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given as $m = \frac{y_2 - y_1}{x_2 - x_1}$ OR $m = \frac{y_1 - y_2}{x_1 - x_2}$

2. Slope $'m'$ of line making angle θ, with x-axis is given as $m = tan\theta$

3. Slope $'m'$ of line $ax + by + c = 0$, is $m = -\frac{co\text{−}efficient\ of\ x}{co\text{−}efficient\ of\ y} = -\frac{a}{b}$

Remarks:

- Slope of horizontal line or x-axis is always zero i.e. $m = 0$
- Slope of vertical line or y-axis is always undefined i.e. $m = \infty$
- If two lines having slopes m_1 and m_2 are parallel then $m_1 = m_2$
- If two lines having slopes m_1 and m_2 are parallel then $m_1 m_2 = -1$, this case is not applicable for co-ordinates axes

- Angle of intersection between lines having slopes m_1 and m_2 is $\mathbf{tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}}$
- Two lines $a_1 x + b_1 y + c_1 = 0$ and $a_2 x + b_2 y + c_2 = 0$ are not solvable or has no unique solution if
$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

POINT OF INTERSECTION OF TWO LINES

The point of intersection b\w two lines $a_1 x + b_1 y + c_1 = 0, a_2 x + b_2 y + c_2 = 0$ is given by

$$\frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}} = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

Equation of Line

1. General equation of line in plane is $ax + by + c = 0$
2. Symmetric form of line joining points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is $\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1}$
3. Determinant form of line joining $P(x_1, y_1)$ and $Q(x_2, y_2)$ is
$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$
4. Equation of line through point $P(x_1, y_1)$ having slope $'m'$ is
$$y - y_1 = m(x - x_1)$$
5. Slope intercept form of line is $y = m\,x + c$
6. Two-intercept form is $\frac{x}{a} + \frac{y}{b} = 1$, a & b are respectively known as x-intercept and y-intercept
7. Normal form of line $ax + by + c = 0$ is given as $cos\alpha\, x + sin\alpha\, y = p$,
   Where $cos\alpha = \frac{a}{\sqrt{a^2 + b^2}}$, $sin\alpha = \frac{b}{\sqrt{a^2 + b^2}}$ and $p = \frac{-c}{\sqrt{a^2 + b^2}}$

Co-linear Points

If points $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ lie on same line then they will be called as co-linear points. In this case

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0 \quad \text{Or} \quad \frac{x_3 - x_1}{x_2 - x_1} = \frac{y_3 - y_1}{y_2 - y_1}$$

Area of Triangle

Area of triangle having vertices $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ is given as

$$\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Concurrent Lines

Three or more lines intersecting (meeting) at one point are called concurrent lines. If three lines

$$a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0 \;\&\; a_3x + b_3y + c_3 = 0$$

Are concurrent then

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Pair of Lines through origin

Pair of lines through origin is given as $ax^2 + 2hxy + by^2 = 0$

Angle of intersection between two lines in this case is given as $\tan\theta = \frac{2\sqrt{h^2 - ab}}{a+b}$

Note: Lines will be real if $h^2 - ab \geq 0$ and will be coincident if $h^2 - ab = 0$

Some important terms in plane Geometry

- A plane polygon having four sides is called quadrilateral
- A quadrilateral having opposite sides as equal and parallel is called a parallelogram
  - (i). If ABCD is parallelogram then the following holds $AB = CD \text{ and } AD = BC$
  - (ii). If ABCD is parallelogram then the following holds

    $$slope\ of\ AB = slope\ of\ CD\ \ and\ \ slope\ of\ AD = slope\ of\ BC$$
  - (iii). Diagonals of a parallelogram are always bisect.

- A parallelogram whose all interior angles are 90° is called rectangle. In rectangle diagonals are always equal in length
- A parallelogram whose all sides are equal in length is called rhombus. In rhombus diagonals are not equal in length but always perpendicular to each other
- A square is a rhombus whose all interior angles are 90° and diagonals equal in length
- A triangle having one of its angles is of 90° is called a right angle. In right angle base and perpendicular must be at right angle to each other and following holds $(HYP.)^2 = (BASE)^2 + (PERP.)^2$
- A triangle having two sides of equal in length is called an isosceles triangle
- A triangle having all sides equal in length is called equilateral triangle and each interior angle is 60°

CHAPTER 6 CONIC SECTIONS

Circle: Set of points in plane whose distance from a fixed point is always constant.

Fixed point is called center while distance is called radius

Mathematically

$$A = \{P(x,y): (x-h)^2 + (y-k)^2 = r^2\}$$

General equation of circle is, $x^2 + y^2 + 2gx + 2fy + c = 0$

With center: $C(-g,-f)$ and radius: $r = \sqrt{g^2 + f^2 - c}$

Standard equation of circle is $(x-h)^2 + (y-k)^2 = r^2$, where $C(h,k)$ is center of circle and r be the radius.

Parabola: Set of all points in plane whose distance from a fixed point and a fixed line is always constant. Fixed point is called focus and fixed line is called directrix. A line through focus and perpendicular to directrix is called Axis of Parabola. Point of intersection of parabola and axis of parabola is known as vertex. A chord through focus of parabola is called focal chord. A chord through focus and perpendicular to axis of parabola is called latusrectum. For parabola $y^2 = 4ax$, length of latusrectum is $|4a|$

Standard Equation of Parabola

$(y-k)^2 = 4a(x-h)$	$(x-h)^2 = 4a(y-k)$
Vertex: $V(h,k)$	Vertex: $V(h,k)$
Axis: Since degree of x is one so axis parallel to x-axis.	Axis: Since degree of y is one so axis parallel to y-axis.
Focus: $F(h+a,k)$	Focus: $F(h,k+a)$
Directrix: $x = h - a$	Directrix: $y = k - a$

Eccentricity of parabola is always '1'

Ellipse:

Set of all points in plane ratio of whose distance from a fixed point and a fixed line bears a constant ratio, i.e. the fixed point is called focus and fixed line is called directrix.

In case of ellipse there exist two foci and two directricies. If F be one of the foci and P is any point on the ellipse, distance from P to F is $|PF|$ and distance from P to corresponding directrix is $|PM|$ then

$\frac{|PF|}{|PM|} = e < 1$ $\quad where\ 'e'$ Is constant and known as eccentricity

General Equation of Ellipse

$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$
Center: $C(h,k)$	Center: $C(h,k)$
Major Axis: $\lvert a\rvert > \lvert b\rvert$, then major axis parallel to x- axis	Major Axis: $\lvert a\rvert < \lvert b\rvert$, then major axis parallel to y-axis
Vertices: $A(h \pm a, k)$	Vertices: $A(h, k \pm b)$
Co-vertices: $B(h, k \pm b)$	Co-vertices: $B(h \pm a, k)$
Foci: $F(h \pm ae, k)$, where $b^2 = a^2(1-e^2)$	Foci: $F(h, k \pm be)$, where $a^2 = b^2(1-e^2)$
Directrices: $x = h \pm \frac{a}{e}$	Directrices: $y = k \pm \frac{b}{e}$

Hyperbola:

Set of all points in plane ratio of whose distances from a fixed point and a fixed line bears a constant ratio, i.e. the fixed point is called focus and fixed line is called directrix.

In case of ellipse there exist two foci and two directricies. If F be one of the foci and P is any point on the ellipse, distance from P to F is $|PF|$ and distance from P to corresponding directrix is $|PM|$ then

$\frac{|PF|}{|PM|} = e > 1 \qquad where\ 'e'$ Is constant and known as eccentricity

General Equation of Hyperbola

$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = -1$
Center: $C(h,k)$	Center: $C(h,k)$
Major Axis: The major axis parallel to x- axis	Major Axis: The major axis parallel to y-axis
Vertices: $A(h \pm a, k)$	Vertices: $A(h, k \pm b)$
Co-vertices: $B(h, k \pm b)$	Co-vertices: $B(h \pm a, k)$
Foci: $F(h \pm ae, k)$, where $b^2 = a^2(e^2 - 1)$	Foci: $F(h, k \pm be)$, where $a^2 = b^2(e^2 - 1)$
Directrices: $x = h \pm \frac{a}{e}$	Directrices: $y = k \pm \frac{b}{e}$

General Equation of conic in Plane

$$\boldsymbol{a\,x^2 + 2hxy + by^2 + 2gx + 2fy + c = 0}$$

1. If $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$, then conic represents a pair of straight lines
2. If $h^2 - ab = 0$, then conic is parabola
3. If $h^2 - ab > 0$, then conic is hyperbola
4. If $h^2 - ab < 0$, then conic is ellipse

Angle of rotation for axis of conic is $tan2\theta = \frac{2h}{a-b}$, where $\theta < 90°$

Transformation equations for axes of rotation are $x = X\cos\theta - Y\sin\theta,\ y = X\sin\theta + Y\cos\theta$

Remarks:

- If $h = 0$ then axis of conic will be parallel to co-ordinate axes
- If $h = 0, g = 0,\ f = 0$ & $c = 1$ then axes of conic coincides with co-ordinates axes
- If $c = 0$, then conic passes through origin
- If $a = 0, b = 0$ & $h = 0$, then conic will be a straight line
- If $h = 0$ & $a = b$ then conic will be a circle
- Equation of circle having center at origin $C(0,0)$ and radius r is $x^2 + y^2 = r^2$
- A circle whose radius is zero is called a point circle.
- Standard equations of parabola are $y^2 = 4ax$, $x^2 = 4ay$ if $a > 0$ then parabola opens right, upward resp. if $a < 0$ then parabola opens towards left, down resp.
- Standard equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
- Standard equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- Eccentricity ‘e’ of parabola is one (e=1), of ellipse less than one (e<1) and for hyperbola greater than one (e>1).

MPORTANT NOTE: (i). CIRCLE: The set of all points in plane whose distance from a fixed point is the same, fixed point is called center and fixed distance is called radius

(ii). PARABOLA: The set of all the points in plane whose distance from a fixed point and a fixed line is constant, fixed line is called directrix & fixed point is called focus.

(a). a line passing through focus of parabola and perpendicular to the directrix is called AXIS of parabola

(b). The point of intersection of parabola and axis of parabola is called vertex of parabola '' note that the distance b/w vertex to directrix and vertex to focus is the same

(c). LATUSRECTUM: A perpendicular focal chord is called latusrectum. For parabola,

length of latusrectum $= |4a|$

(iii). TANGENT: A line which touches any curve only at its one point is called tangent to that curve

(iv). NORMAL: A line perpendicular to the tangent and passing through the point of tangency. Point of tangency is the common point b/w conic and the tangent.

(v). CHORD: A line segments whose end points lie on any curve

A chord which passes through the center of a circle is called diameter

(vi). HYPERBOLA: Set of all points in plane whose distance from a fixed point & a fixed line gives constant ratio whose value is greater than 1 & this constant ratio is called eccentricity denoted by 'e'

(a). fixed point is known as focus, a hyperbola has two foci and two fixed lines known as directricies

(b). MAJOR AXIS: A line perpendicular to directricies and passing through foci is called major axis

(c). CENTER: the midpoint of foci is called center. A line perpendicular to axis of hyperbola & passing through the center of hyperbola is called minor axis.

ELLIPSE: Set of all points in plane whose distance from a fixed point & a fixed line gives constant ratio whose value is less than 1 & this constant ratio is called eccentricity denoted by 'e'

(a). fixed point is known as focus, an ellipse has two foci and two fixed lines known as directricies

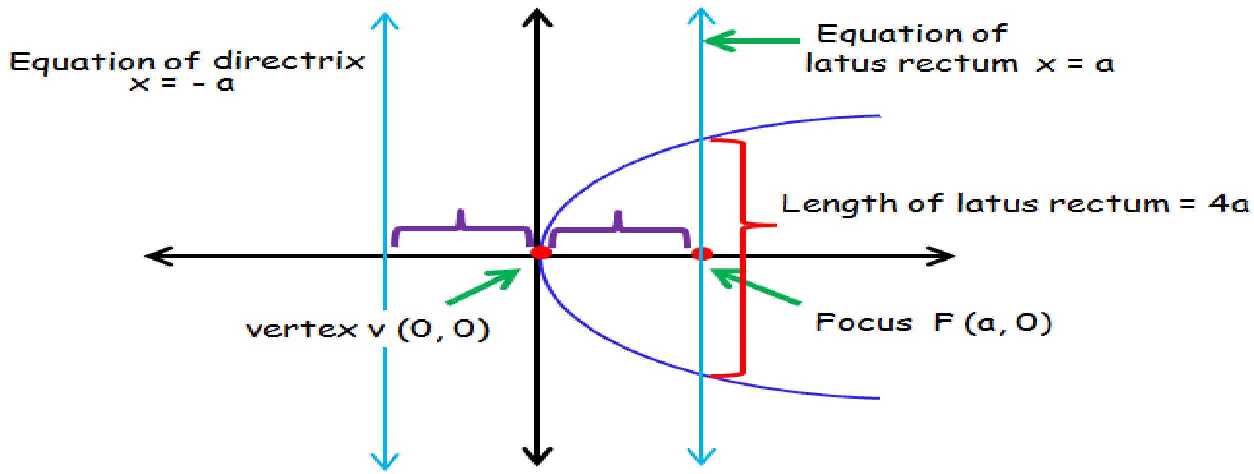
(b). MAJOR AXIS: A line perpendicular to directricies and passing through foci is called major axis

(c). CENTER: The midpoint of foci is called center. A line perpendicular to axis of an ellipse & passing through the center of ellipse is called minor axis.

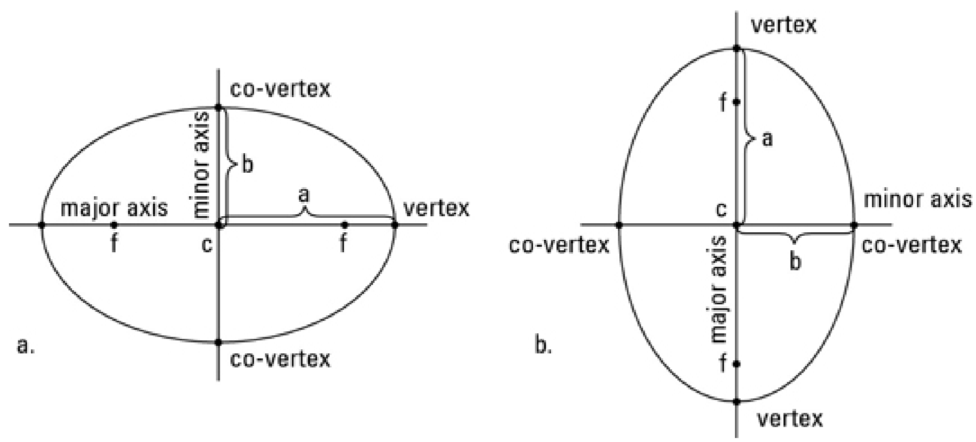
General Equation of Conic in Polar Form			
Axis (Major axis) along x-axis		Axis (Major axis) along y-axis	
$r = \dfrac{l}{1 \pm e\cos\theta}$		$r = \dfrac{l}{1 \pm e\sin\theta}$	
Parabola	Ellipse	Hyperbola	Circle
$e = 1$	$e < 1$	$e > 1$	$e = 0$

Note: *note that* $l = ed$, where d is distance between focus and directrix, if the axes do not coincide with coordinate axes, then conic becomes $\boldsymbol{r = \frac{l}{1 \pm e\cos(\theta-\theta_o)}}$ $\quad \boldsymbol{or} \quad$ $\boldsymbol{r = \frac{l}{1 \pm e\sin(\theta-\theta_o)}}$, where θ_o is angle between new axis and x-axis

Parabola $(y^2 = 4ax)$



Ellipse $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1\right)$



Hyperbola $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\right)$